



Osprey

Distributed Control Systems

Machine Learning Data Platform (MLDP) Overview

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Machine Learning Data Platform

Overview

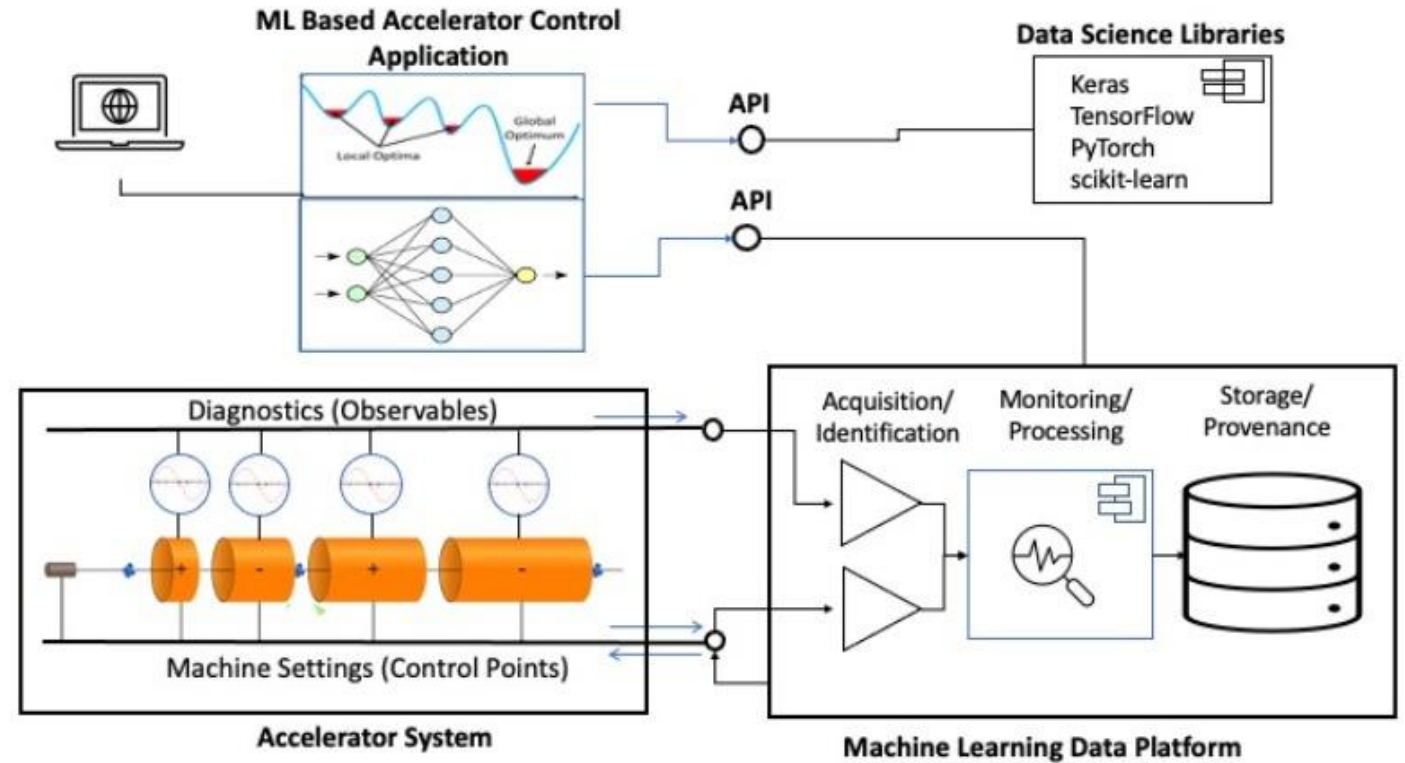
APIs

Ecosystem

Development Plans

Learn More

The MLDP provides tools for 1) building an annotated archive of accelerator PV time-series data and 2) using that archive to build and operate data science applications.



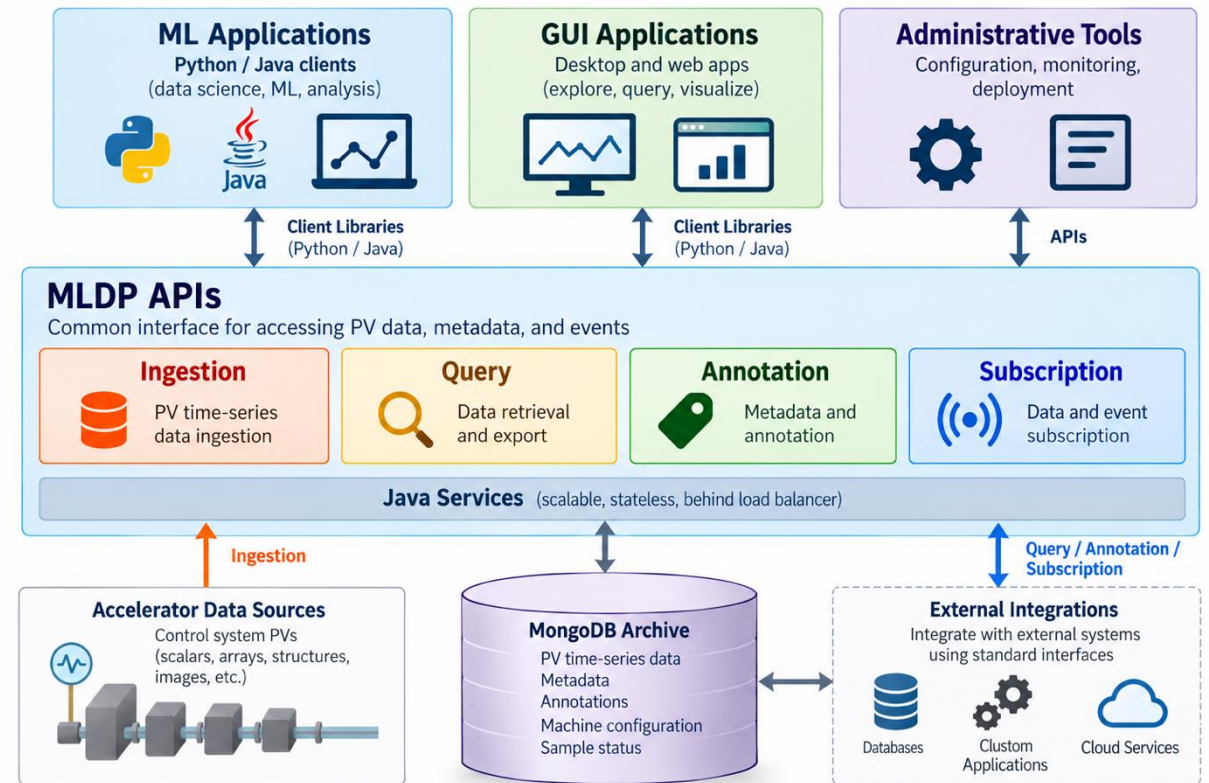
The MLDP defines a set of APIs and includes an ecosystem of components communicating via those APIs.

APIs:

- PV time-series data ingestion
- metadata and annotation
- data retrieval
- data and event subscription

Ecosystem Components:

- scalable Java services managing a MongoDB data archive
- Python and Java API clients for building applications
- desktop and web GUI apps
- administrative tools and configuration examples



MLDP APIs

Ingestion

Annotation

Query

Subscription

PV Time-Series Data Ingestion API

- Provides optimized structures for ingesting a variety of data types including scalars, multi-dimensional arrays, structures, images, and serialized data
- Utilizes streaming API methods and supports static and dynamic load balancer configurations for maximum performance
- Includes mechanisms for specifying data provenance and ingestion metadata
- Supports both continuous and batch ingestion

MongoDB Archive – Particle Accelerator Data

← Process Variables (Columns) →

	PV1	PV2	PV3	PV4
Timestamp	Beam Energy (GEV)	Magnet Field (T)	RF Cavity Voltage (MV)	Beam Current (ma)
	PV1	PV2	PV3	PV4
2023-07-15 12:00:00	4.20	1.50	12.5	250.4
2023-07-15 12:01:00	4.18	1.55	12.7	248.9
2023-07-15 12:01:00	4.22	1.52	12.5	251.1
2023-07-15 12:03:00	4.19	1.54	12.6	249.7
2023-07-15 12:04:00	4.21	1.56	12.8	252.0

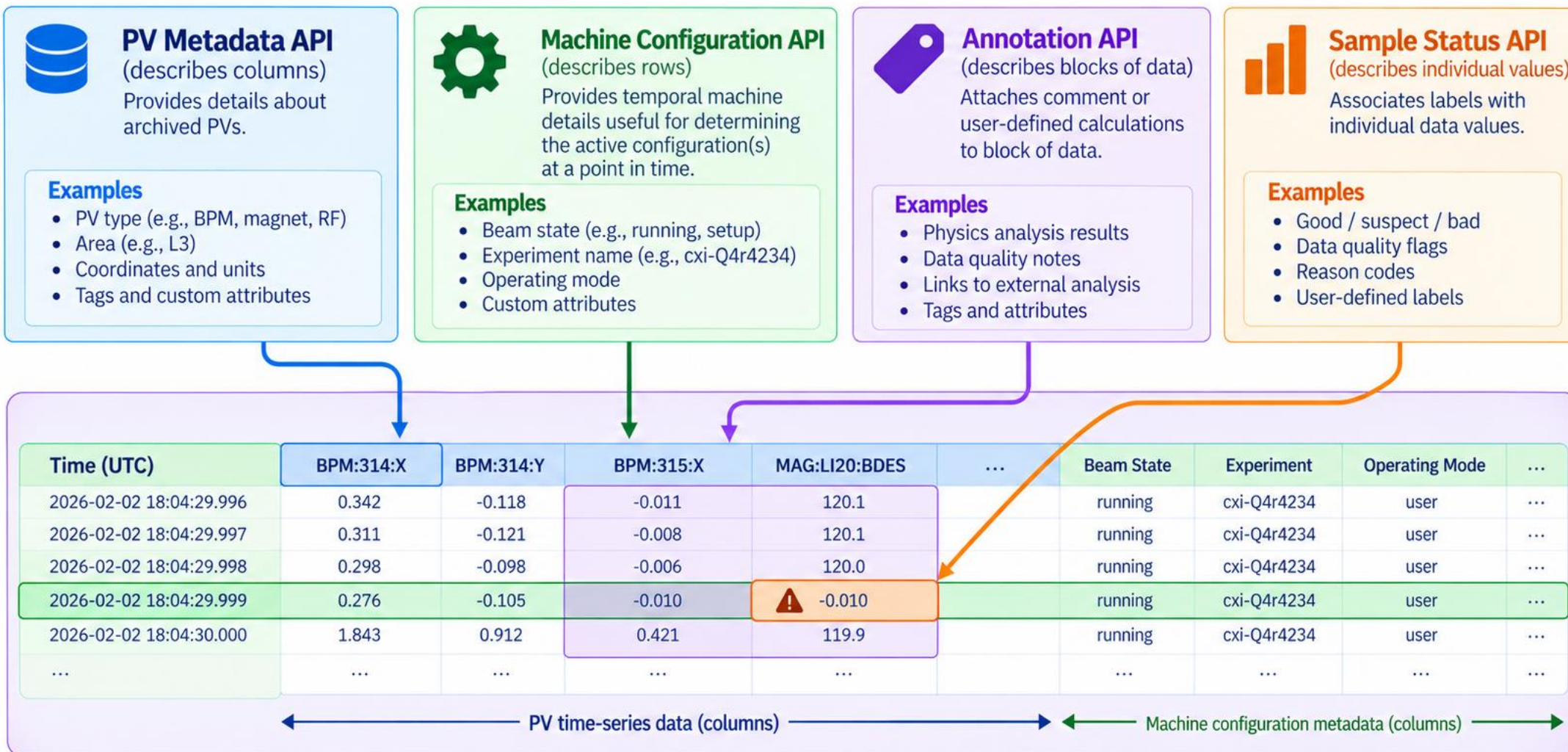
← PV Data Values →

Timestamps (Rows)

Time Samples

MLDP Archive Metadata

A unified store for time-series data with rich metadata that describes the columns, rows, and individual values.



Querying Time-Series Data

Use metadata, machine configuration, and sample status to retrieve the right data, without knowing PV names or timestamps.



Get me all **good BPM data** from **area L3** when we were **running beam** and the experiment was **cxi-Q4r4234**.



What data?

PV Metadata

BPMs in L3

Select PVs using metadata
(e.g., type = BPM, area = L3).



When?

Machine Configuration

Running beam
Experiment = cxi-Q4r4234

Use machine configuration metadata
to find the relevant time period(s).



Which values?

Sample Status

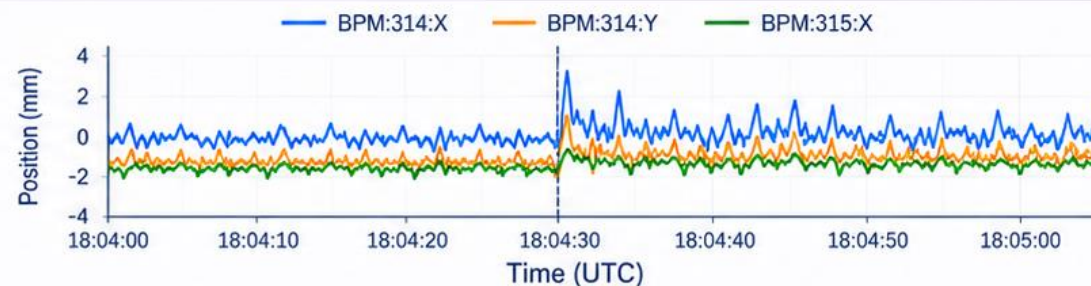
Good data only

Filter by sample status to include
only good data.

MLDP Query

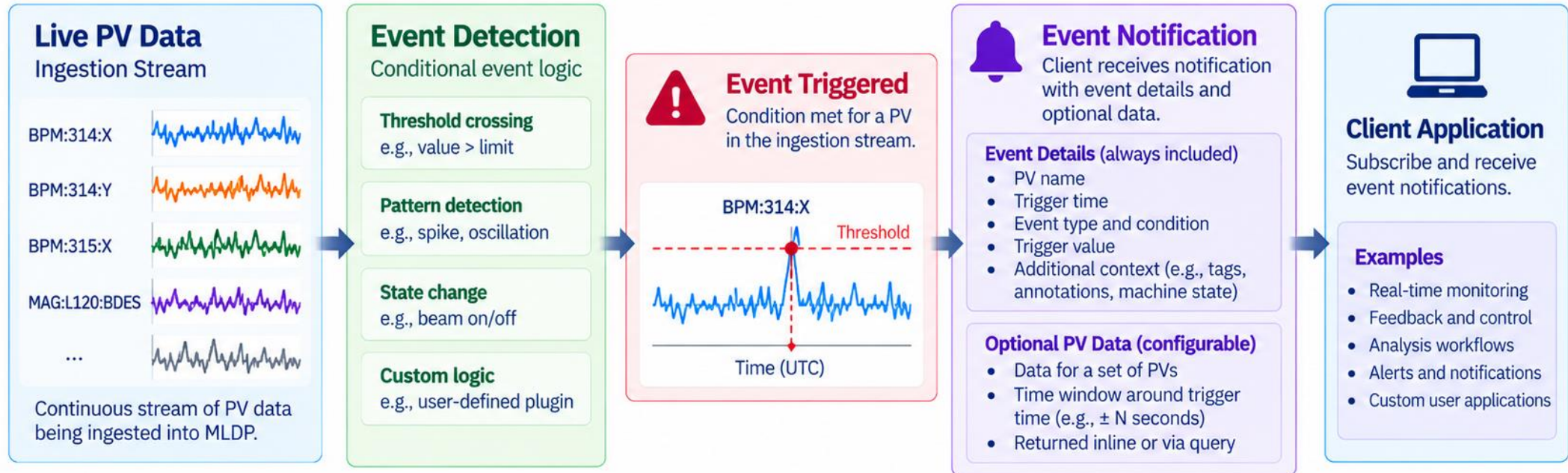
Matching Time-Series Data

Time (UTC)	BPM:314:X (mm)	BPM:314:Y (mm)	BPM:315:X (mm)
2026-02-02 18:04:29.996	0.342	-0.118	-0.011
2026-02-02 18:04:29.997	0.311	-0.121	-0.008
2026-02-02 18:04:29.998	0.298	-0.098	-0.006
2026-02-02 18:04:29.999	0.276	-0.105	-0.010
2026-02-02 18:04:30.000	1.843	0.912	0.421



Data Event Subscription API

Subscribe to live PV data and receive notifications when conditional events are triggered, optionally including data for a set of PVs and a window of time around the trigger.



Future Extensions (Under Investigation)



User-Defined Plugins

- Run custom event detection logic as a plugin.
- Implement domain-specific algorithms.
- Deploy and manage plugins with MLDP services.



Publish to External Frameworks

- Publish event notifications and/or data to frameworks like Kafka.
- Enable integration with external analysis and control systems.
- Support flexible routing and filtering of events.

MLDP Ecosystem

Java Server Applications

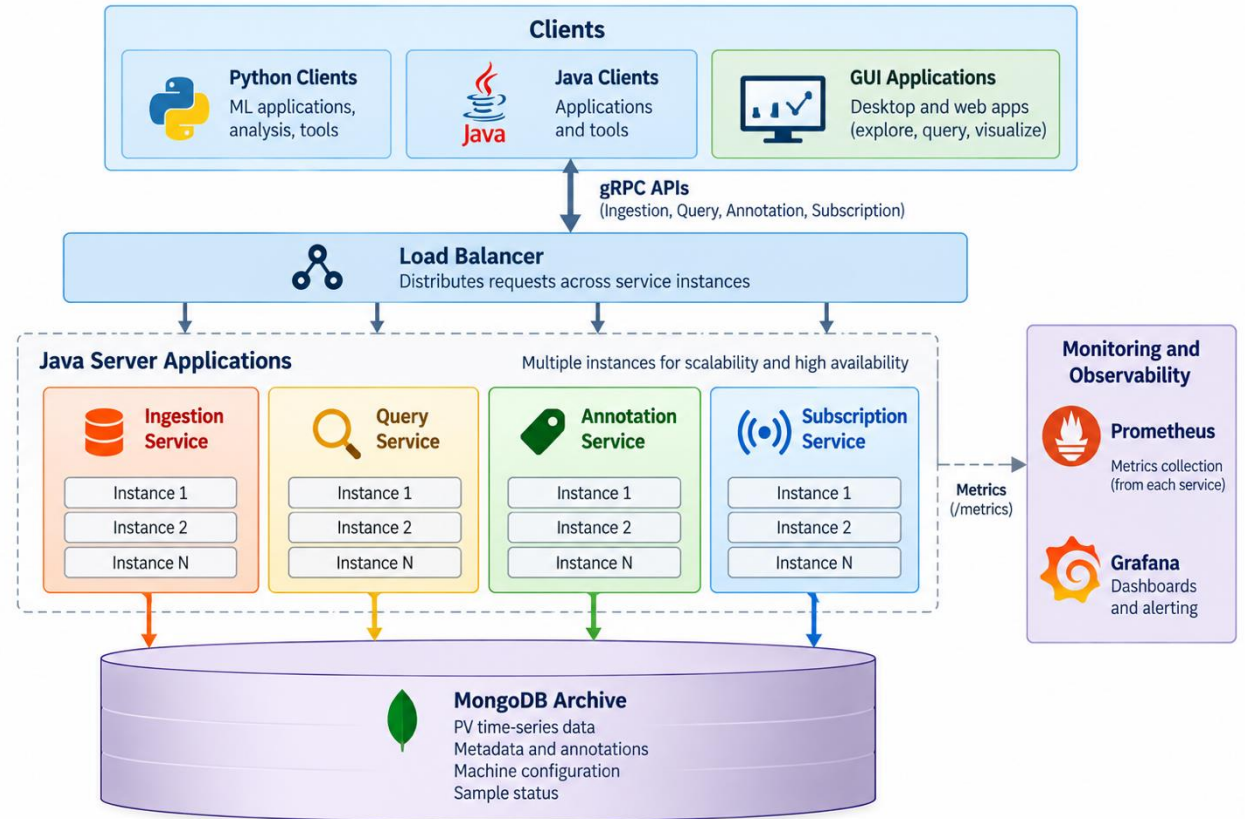
Client API Libraries

GUI Applications

System Administration Tools

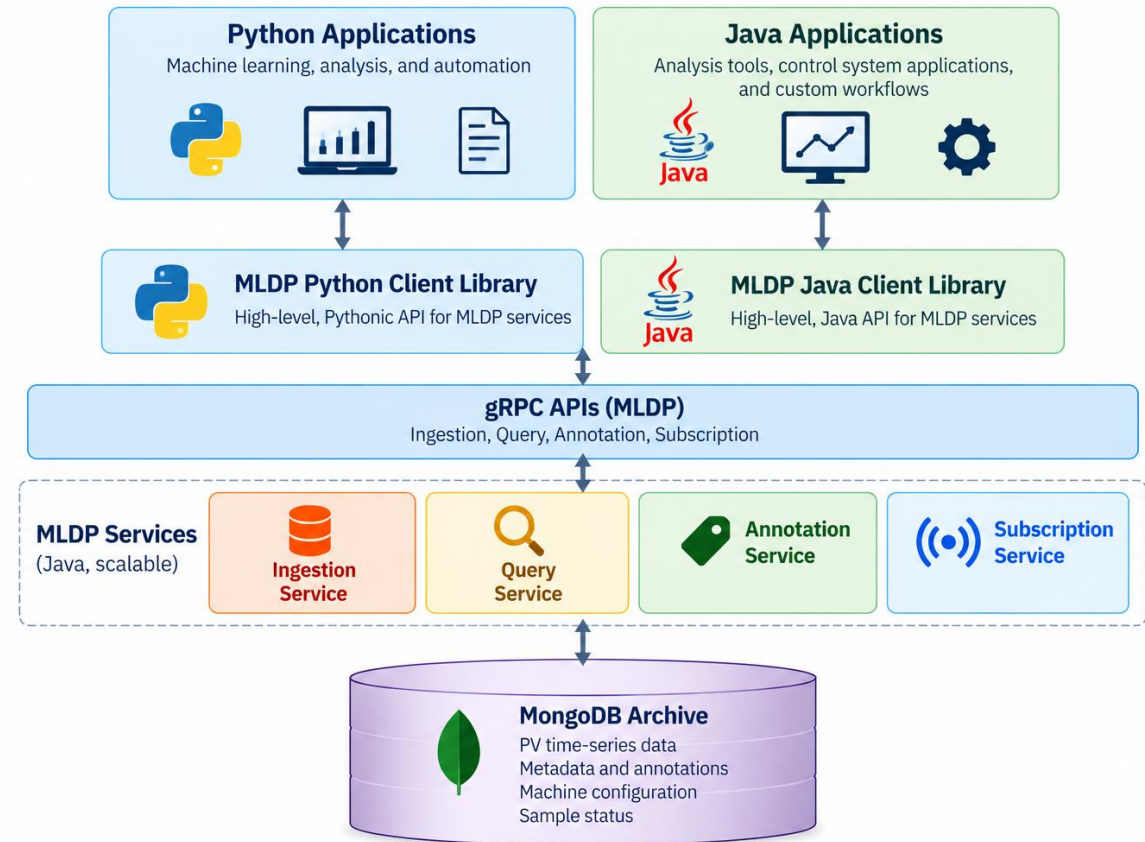
Java Server Applications

- Each API is handled by a Java server application.
- A load balancer is used to run multiple server instances in parallel for maximum performance.
- Support is provided for advanced MongoDB configurations.
- Each service exports metrics over a Prometheus scrape endpoint to facilitate monitoring.
- Migration framework manages evolution of database schema for new releases.



Client API Libraries

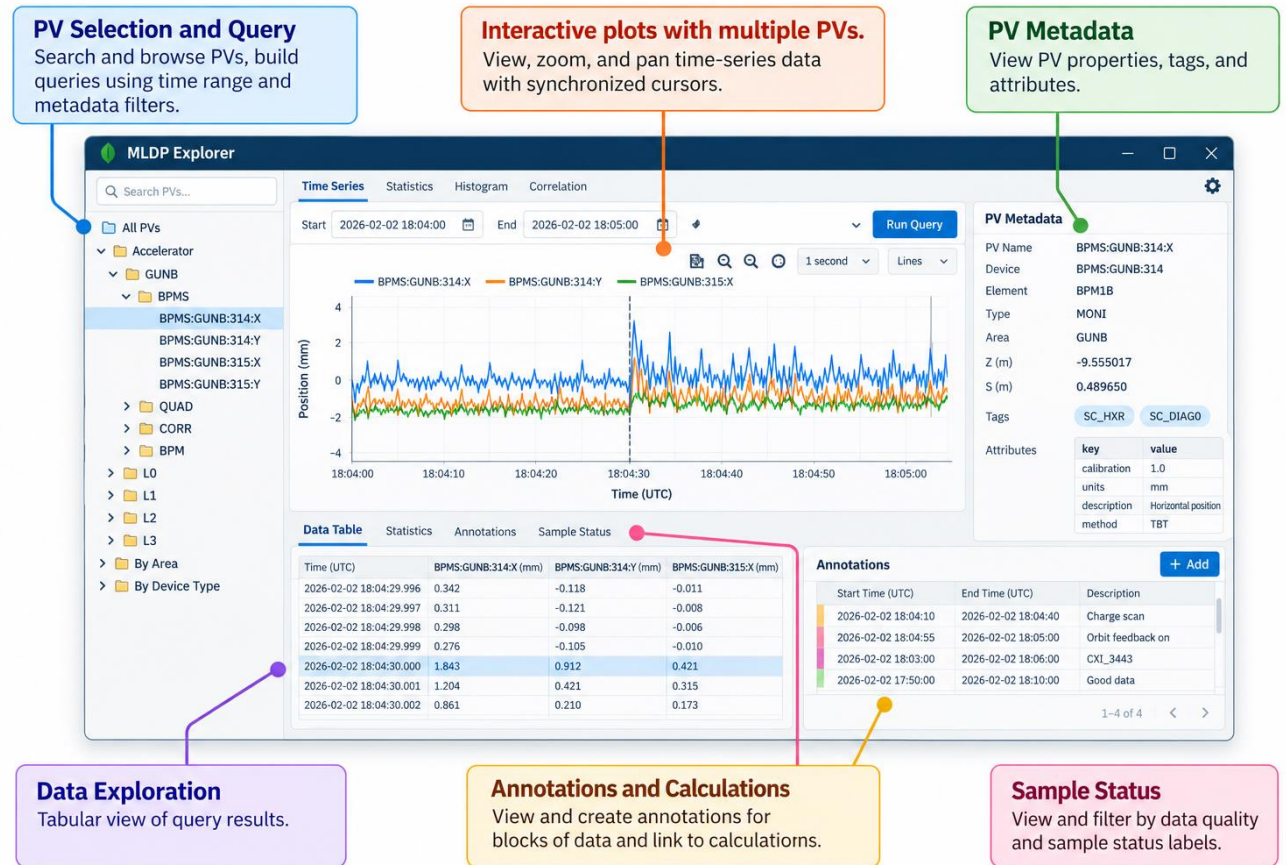
- A Python client library is under development
- The initial library implementation provides wrappers for calling the MLDP APIs.
- Current development focus is to add higher-level “Pythonic” support for data science applications.
- Two Java client libraries are under development.
 - ✓ High-level library provides utilities for both ingestion and query of PV time-series data, but needs “modernization”.
 - ✓ Low-level library is used as foundation of desktop GUI application and can be extended for other client uses.



➤ Python cookbook documentation <https://github.com/osprey-dcs/dp-python-lib/tree/main/doc/cookbook>

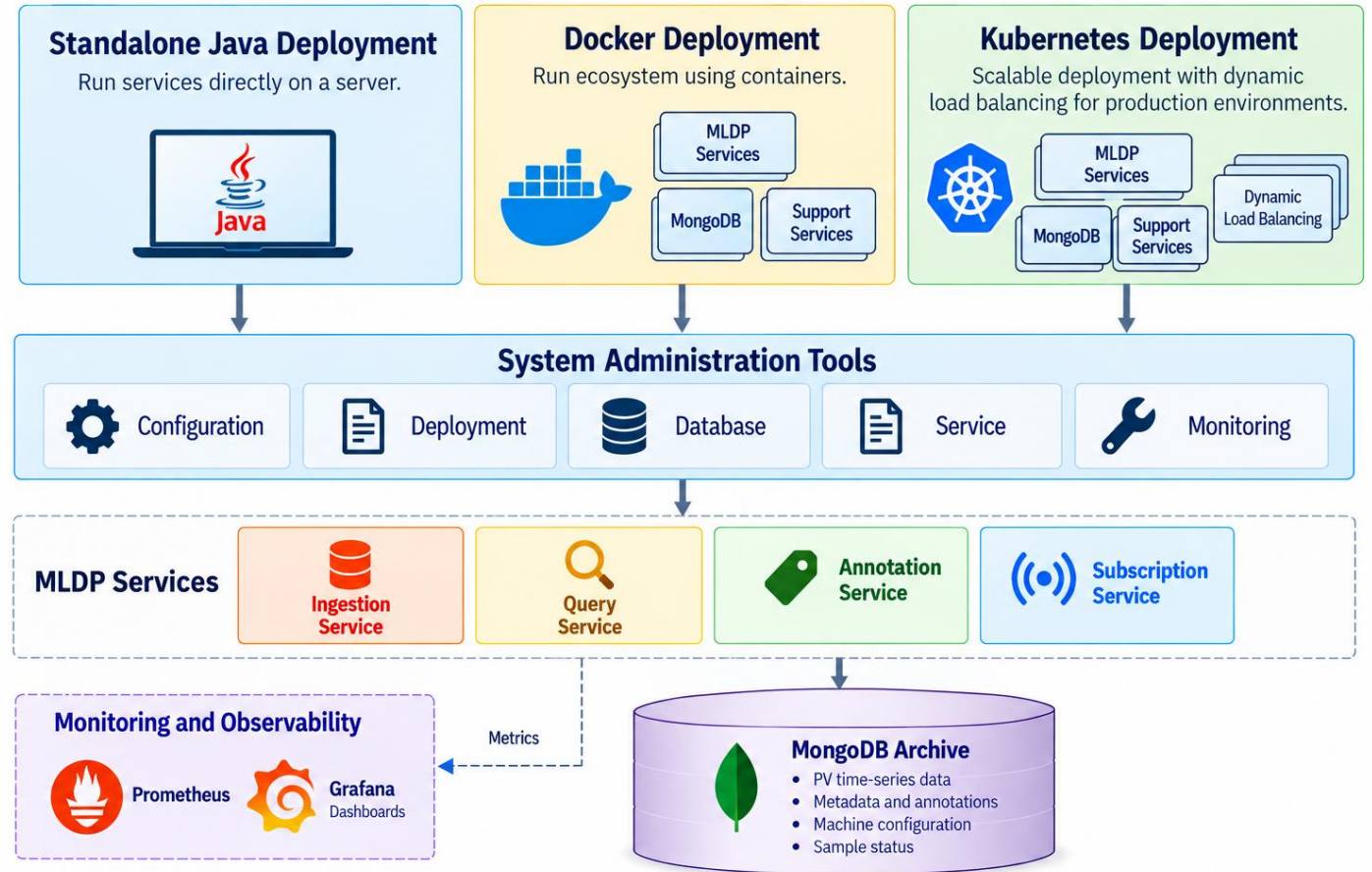
GUI Applications

- A Java desktop GUI application was built to support demonstration of the MLDP APIs but designed to be of general use.
- Provides views and controls for:
 - Ingesting tabular data files
 - Subscribing to events from the data ingestion stream
 - Creating PV and machine configuration metadata
 - Creating annotations and uploading calculations
 - Exploring the archive across several dimensions including PV metadata, machine configurations / activations, sample status, and annotations
 - Building PV time-series data queries with filtering by metadata criteria
 - Viewing PV time-series data
- A prototype JavaScript web application was built but needs work to support full set of MLDP features.



System Administration Tools

- The master project repository includes tools for installing, configuring, and operating the MLDP ecosystem.
- Provides utilities and examples for deployments using standalone Java applications, Docker containers, Envoy static load balancing, and Kubernetes dynamic load balancing.
- Includes overview documentation with project history, roadmap, and links to child repositories.



Project Details

Development Plans

For More Information

MLDP Development Plans

Support for SLAC high-repetition Superconducting (kHz) and High Energy accelerator upgrades

- ✓ API and database schema optimizations
- ✓ More efficient use of storage via tiering and offline indexing
- ✓ strategies for improved distribution of data in Mongo shards

Other FY27 development priorities

- ✓ Python client library data science enhancements
- ✓ Improvements to data subscription frameworks
- ✓ Support for large atomic data values (images)
- ✓ Tools for ingesting data from EPICS environment

TODO / looking for funding

- ☐ JWT authentication / role-based authorization / LDAP integration
- ☐ Data ownership and sharing mechanisms
- ☐ Java client library enhancements

For Additional Information

The primary GitHub repo for the Machine Learning Data Platform project is <https://github.com/osprey-dcs/data-platform> . It includes project overview, status, road map, quick start guide, installation instructions, and links to secondary repos and additional project documents.

Project Team Contacts

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API and Client library cookbook documentation

- <https://github.com/osprey-dcs/dp-grpc/tree/main/doc/cookbook>
- <https://github.com/osprey-dcs/dp-python-lib/tree/main/doc/cookbook>